

A 3D moisture-stress FEM analysis for time dependent problems in timber structures

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Abstract This paper presents a 3D moisture-stress numerical analysis for timber structures under variable humidity and load conditions. An orthotropic viscoelastic-mechanosorptive material model is specialized on the basis of previous models. Both the constitutive model and the equations needed to describe the moisture flow across the structure are implemented into user subroutines of the Abaqus finite element code and a coupled moisture-stress analysis is performed for several types of mechanical loads and moisture changes. The presented computational approach is validated by analyzing some wood tests described in the literature and comparing the computational results with the reported experimental data.

Keywords Timber structures · Moisture induced stresses · Creep models · Abaqus code

1 Introduction

The mechanical response of wood in presence of moisture changes is a fundamental topic for both the serviceability and the safety of timber structures (Toratti and Svensson 2000; Hanhijärvi and Mackenzie-Helnwein 2003; Chassagne et al. 2006). It is assumed that stresses in the plane perpendicular to the grain direction affect the strength of glulam beams during variable environmental conditions (Ormarsson 1999; Jönsson 2005). Also the strength of timber connections in the cross grain direction is influenced by moisture induced deformations (Sjödén 2004). In general, the combination of moisture history and mechanical loading is important under service conditions, that is under relative low loads and natural temperatures and humidities (Toratti and Svensson 2002). According to Ranta-Maunus (2003), in this context the moisture effect can be considered as an action to be combined with the external mechanical loads. The failure modes in wood and the crack growth under both constant and varying humidity levels in timber structures are challenging research topics as well (Mackenzie-Helnwein et al. 2003; Schmidt and Kaliske 2009; Liyu et al. 2003).

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In the last two decades, several material models based on experiments under different mechanical loads and humidity conditions were proposed (see references in Hanhijärvi 1995; Vidal-Sallé and Chassagne 2007). These models take into account the interaction between varying moisture and stress, called mechanosorption or mechanosorptive creep, and sometimes neglect the effect of the viscoelastic creep, which is normally found at constant moisture contents. Hanhijärvi and Mackenzie-Helnwein (2003) proposed a general 3D mathematical formulation for creep of orthotropic wood in drying conditions where the coupling between viscoelastic and mechanosorptive creep was taken into account. However, only a 2D mechanical analysis of wood in the section perpendicular to grain was performed. In fact, since wood is a variable, heterogeneous and anisotropic material, the identification of creep parameters for 3D models is a very difficult task. In spite of the available experimental data on the viscoelastic creep and the mechanosorptive effect in timber structures in grain or in cross grain direction, there is still a lack of both experimental work and theoretical background, for example about the Poisson ratios to be used in 3D wood material models in the presence of viscoelastic effects and moisture gradients. Recent advances on this subject were presented by Frandsen (2007). On the other hand, for the efficient design of timber structures, the formulation of a reliable 3D model for wood is important and the finite element code Abaqus represents a suitable computational tool for implementing constitutive models and for the simulation of three-dimensional coupled problems in engineering (Abaqus 2004a). Recent literature shows a growing interest in modelling 3D orthotropic wood structures by Abaqus (Ormarrsson 1999; Sjödin 2004; Vidal-Sallé and Chassagne 2007). Also 3D fracture mechanics problems in wood can be solved, at least in the elastic case, by using the cited code (see de Moura et al. 2006; Vasic and Stanzl-Tschegg 2007).

This paper presents a 3D coupled moisture-stress numerical analysis for timber structures designed to be used in service conditions of buildings. The analysis is based on a constitutive orthotropic viscoelastic-mechanosorptive model. The rheological model is characterized by five deformation mechanisms and, as proposed by Hanhijärvi and Mackenzie-Helnwein (2003), a thermodynamic formulation is used starting from the Helmholtz free energy expressed as a function of temperature, moisture content, total strain, viscoelastic strain and mechanosorptive strain. The used viscoelastic model is an extension of the 1D formulation proposed by Toratti (1992) for parallel to grain direction and consists of a sum of Kelvin type elemental deformations. The mechanosorptive model contains an irrecoverable part plus a series of Kelvin elements and is a combination of the 1D model introduced by Toratti and Svensson (2002) for cross grain direction and of the 1D model for longitudinal direction of wood presented by Toratti (1992). The extension of the previous models to 3D is based on three-dimensional elemental viscoelastic and mechanosorptive matrices. In this work, temperature is considered to be constant. The temperature effect was not found of relative significance on the deformation behaviour, at least in the natural temperature ranges (Toratti 1992).

The viscoelastic-mechanosorptive constitutive model is implemented in the user subroutine UMAT of the FEM code Abaqus (Abaqus 2004b). The algorithm for moisture induced stress update, based on the concept of algorithmic tangent operator (Simo and Hughes 1998), is a variant of the one proposed by Mackenzie-Helnwein and Hanhijärvi (2003). Furthermore, the equations needed to describe the moisture flow across the structure are implemented into the Abaqus user subroutine DFLUX. Using the analogy with the temperature-displacement analysis available in the Abaqus/Standard code, a coupled moisture-stress analysis is performed under different mechanical loads and moisture changes. The analysis is validated by analyzing some of the tests carried out by Leivo (1991), Toratti and Svensson (2000, 2002) and Jönsson (2005) and by comparing the computational results with the

reported experimental data. These comparisons are an important aspect of the present paper. In this work, particular emphasis is posed on the general and flexible 3D computational tool which permits to successfully solve coupled moisture-stress analyses of timber structures under different types of loads and environmental conditions.

2 A 3D orthotropic viscoelastic-mechanosorptive creep model for wood

Wood is usually described as a continuum and homogeneous material with cylindrical orthotropy (Hanhijärvi 1995). This is due to the particular growth mechanism in circular increments which produces the annual rings in the cross section and mainly longitudinally oriented cells (called tracheids at the microscopic level). The anisotropy in the cross sectional plane is due, at least partly, to the growth in radial cell rows which determines unbroken cell wall lines, called wood rays. Furthermore, because of the mainly longitudinally oriented cells, a larger anisotropy between the cross sectional plane and the longitudinal direction is observed. In this work the macroscopic level is defined as the scale of annual ring formation. However, the earlywood and the latewood rings are not considered as different materials. This results in an orthotropic material in which the radial (R) and the tangential (T) axes are oriented in directions where the earlywood and latewood rings are oriented in serial and in parallel directions in the cross section plane. Considering also the grain direction (L), the material at local level is orthotropic. In the Abaqus numerical model used in this paper (see Sect. 4) also the influence of the pith (origin) position has been taken into account by using a cylindrical coordinate system and specifying the material orientation as done in André (2007). The influence of cylindrical coordinate systems on the stress calculation has been discussed for example in Aicher and Dill-Langer (2005). In Fig. 1 the cylindrical coordinates in the plane RT are shown.

To analyze timber structures under service conditions of buildings, the changes of moisture content during loading have to be considered. The 3D orthotropic creep model presented in this work takes into account the hygroexpansion effects and the influence of viscoelastic creep and mechanosorption. Both the viscoelastic and the mechanosorptive creep coefficients presented in the next subsections are obtained by fitting the experimental results in the cross grain and in the grain directions provided in Toratti (1992) and Toratti and Svensson (2000, 2002). These creep parameters are considered to be general for the types of wood analyzed in this study.

2.1 Thermodynamic formulation of the model

According to Sih et al. (1986) and Maugin (1992), thermodynamics is a powerful theoretical tool for the formulation of general problems characterized by the interdependence of heat, moisture and deformation. In the present work temperature is assumed to be constant and moisture transfer is considered to be independent of mechanical variables. In this context,

Fig. 1 Cylindrical coordinates in the plane RT with origin in the pith location

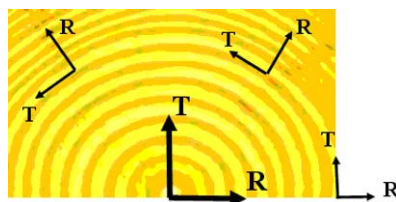
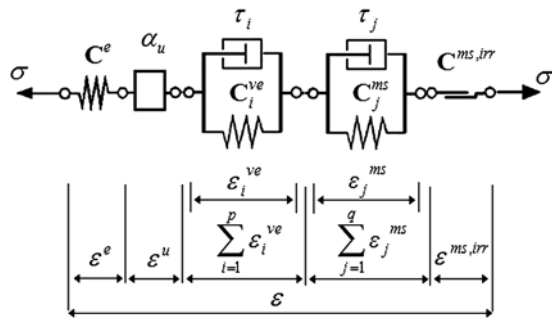


Fig. 2 Scheme of the rheological constitutive model



the main coupling effects are moisture and moisture change-rate dependence of hygroexpansion strain, and viscoelastic and mechanosorptive strain rates, respectively.

From a rheological point of view, the material model presented in this paper is composed of five deformation mechanisms which provide an additive decomposition of strain into elastic response, hygroexpansion, viscoelastic creep, recoverable mechanosorption and mechanosorptive irrecoverable creep (see Fig. 2). Both the viscoelastic and the mechanosorptive recoverable creep are described through Kelvin type elements (Toratti 1992; Toratti and Svensson 2002; Hanhijärvi and Mackenzie-Helnwein 2003). These elements are very suitable for the whole thermodynamic formulation of the problem. The description of irrecoverable strain due to mechanosorption is done by means of the mechanosorptive dashpot scheme proposed in Toratti and Svensson (2002). The total strain tensor $\boldsymbol{\varepsilon}$ of the model is given by

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^u + \sum_{i=1}^p \boldsymbol{\varepsilon}_i^{ve} + \sum_{j=1}^q \boldsymbol{\varepsilon}_j^{ms} + \boldsymbol{\varepsilon}^{ms,irr} \quad (1)$$

where $\boldsymbol{\varepsilon}^e$ is the elastic strain, $\boldsymbol{\varepsilon}^u$ the hygroexpansion strain, $\boldsymbol{\varepsilon}_i^{ve}$ the elemental viscoelastic strain, $\boldsymbol{\varepsilon}_j^{ms}$ the elemental mechanosorptive strain and $\boldsymbol{\varepsilon}^{ms,irr}$ the irrecoverable part of the total mechanosorptive strain. The terms p and q are the number of viscoelastic Kelvin elements and of the mechanosorptive Kelvin elements, respectively.

In this work the thermodynamic formulation of the constitutive model is based on the general approach proposed by Hanhijärvi and Mackenzie-Helnwein (2003). The starting point is the Helmholtz free energy expressed as a function of temperature T , moisture content u , the total strain tensor $\boldsymbol{\varepsilon}$, the elemental strain tensors $\boldsymbol{\varepsilon}_i^{ve}$ and $\boldsymbol{\varepsilon}_j^{ms}$ and the irrecoverable mechanosorptive strain tensor $\boldsymbol{\varepsilon}^{ms,irr}$:

$$\begin{aligned} \Psi(T, u, \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}_i^{ve}, \boldsymbol{\varepsilon}_j^{ms}, \boldsymbol{\varepsilon}^{ms,irr}) &= \Phi(u, T) + \Phi^e(\boldsymbol{\varepsilon}^e) \\ &\quad + \Phi^{ve}(\boldsymbol{\varepsilon}_i^{ve}) + \Phi^{ms}(\boldsymbol{\varepsilon}_j^{ms}) + \Phi^{ms,irr}(\boldsymbol{\varepsilon}^{ms,irr}) \end{aligned}$$

$$\text{with } \boldsymbol{\varepsilon}^e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^u - \sum_{i=1}^p \boldsymbol{\varepsilon}_i^{ve} - \sum_{j=1}^q \boldsymbol{\varepsilon}_j^{ms} - \boldsymbol{\varepsilon}^{ms,irr} \quad (2)$$

where $\Phi(u, T)$ is a general expression for the thermal energy and Φ^e represents the elastic strain energy given by

$$\Phi^e(\boldsymbol{\varepsilon}^e) = \frac{1}{2} \boldsymbol{\varepsilon}^e : \mathbf{C}^e : \boldsymbol{\varepsilon}^e \quad (3)$$

being $\mathbf{C}^e$ the elastic stiffness tensor. The terms $\Phi^{ve}(\boldsymbol{\varepsilon}_i^{ve})$ and $\Phi^{ms}(\boldsymbol{\varepsilon}_j^{ms})$ are the energy stored in the viscoelastic elements and the one stored in the mechanosorptive elements, respectively:

$$\Phi^{ve}(\boldsymbol{\varepsilon}_i^{ve}) = \frac{1}{2} \sum_{i=1}^p \boldsymbol{\varepsilon}_i^{ve} : \mathbf{C}_i^{ve} : \boldsymbol{\varepsilon}_i^{ve}; \quad \Phi^{ms}(\boldsymbol{\varepsilon}_j^{ms}) = \frac{1}{2} \sum_{j=1}^q \boldsymbol{\varepsilon}_j^{ms} : \mathbf{C}_j^{ms} : \boldsymbol{\varepsilon}_j^{ms} \quad (4)$$

where $\mathbf{C}_i^{ve}$ represents the elemental viscoelastic stiffness tensor and $\mathbf{C}_j^{ms}$ the elemental mechanosorptive stiffness tensor. Finally, $\Phi^{ms,irr}(\boldsymbol{\varepsilon}^{ms,irr})$ is the irrecoverable energy related to mechanosorption.

Since this paper presents a finite element analysis, the components of any fourth-order or second-order tensor with respect to a base $\{e_R, e_T, e_L\}$ aligned with axes R – T – L are conveniently rearranged in matrix or vector form following Voigt's notation (Marsden and Hughes 1992; Simo and Hughes 1998; Helnwein 2001). In the next sections the same notations for tensors and matrices are used.

2.2 Elastic deformation

The main constitutive equation of the model is obtained by differentiating the free energy of (2) with respect to the total strain tensor and this provides the total stress tensor

$$\boldsymbol{\sigma} := \frac{\partial \Psi}{\partial \boldsymbol{\varepsilon}} = \mathbf{C}^e : \boldsymbol{\varepsilon}^e \quad (5)$$

In the case of orthotropic materials as wood, the elastic compliance tensor $\mathbf{C}^{e-1}$ in matrix form is

$$\mathbf{C}^{e-1} = \begin{bmatrix} \frac{1}{E_R} & \frac{-\nu_{TR}}{E_T} & \frac{-\nu_{LR}}{E_L} & 0 & 0 & 0 \\ \frac{-\nu_{RT}}{E_R} & \frac{1}{E_T} & \frac{-\nu_{LT}}{E_L} & 0 & 0 & 0 \\ \frac{-\nu_{RL}}{E_R} & \frac{-\nu_{TL}}{E_T} & \frac{1}{E_L} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{RT}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{RL}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{TL}} \end{bmatrix} \quad (6)$$

where the Poisson ratios are related as follows:

$$\nu_{RL} = \frac{E_R}{E_L} \nu_{LR}, \quad \nu_{TL} = \frac{E_T}{E_L} \nu_{LT}, \quad \nu_{TR} = \frac{E_T}{E_R} \nu_{RT} \quad (7)$$

According to Santaoja et al. (1991), both the elastic and the shear moduli are expressed as functions of density, temperature and moisture content:

$$\begin{aligned} E_i &= E_{i,ref}(1 + a_1(\rho - \rho_{ref}) + b_1(T - T_{ref}) + c_1(u - u_{ref})) \\ &\text{with } i = R, T, L \\ G_{ij} &= G_{ij,ref}(1 + a_1(\rho - \rho_{ref}) + b_1(T - T_{ref}) + c_1(u - u_{ref})) \\ &\text{with } i, j = R, T, L; i \neq j \end{aligned} \quad (8)$$

$E_{i,ref}$ and $G_{ij,ref}$ being the values of moduli in reference conditions of density ρ_{ref} , temperature T_{ref} and moisture content u_{ref} . In the examples analyzed in this paper, the density is

considered to be constant ($\rho = \rho_{ref} = 550 \text{ kg/m}^3$ for pine wood and $\rho = \rho_{ref} = 450 \text{ kg/m}^3$ for spruce wood). Also the temperature is assumed to be constant ($T = T_{ref} = 20^\circ\text{C}$). Following (Hanhijärvi 1995), the moisture content $0 < u < 1$ is expressed as $u = (m - m_0)/m_0$ being m the mass of the specimen and m_0 the mass of the specimen in absolute dry condition. The reference value of moisture content is $u_{ref} = 0.12$. The values of the dimensionless parameters in (8) are $a_1 = 0.0003$, $b_1 = -0.007$, $c_1 = 2.6$ (Santaoja et al. 1991).

Referring to (5), the time derivative of the elastic strain is

$$\dot{\boldsymbol{\epsilon}}^e = \dot{\mathbf{C}}^{e-1} : \boldsymbol{\sigma} + \mathbf{C}^{e-1} : \dot{\boldsymbol{\sigma}} \quad (9)$$

where $\dot{\mathbf{C}}^{e-1}$ represents the elastic compliance tensor rate.

2.3 Hygroexpansion

The hygroexpansion strain has the following mathematical rate form:

$$\dot{\boldsymbol{\epsilon}}^u = \boldsymbol{\alpha}_u \dot{u} \quad (10)$$

where $\dot{u}$ represents the moisture content rate and $\boldsymbol{\alpha}_u$ is the second-order hygroexpansion tensor which, rewritten in vector form, is $\boldsymbol{\alpha}_u = [\alpha_R, \alpha_T, \alpha_L, 0, 0, 0]^T$. The values of the hygroexpansion coefficients used in this work are those reported in Santaoja et al. (1991):

$$\alpha_R = 0.13; \quad \alpha_T = 0.27; \quad \alpha_L = 0.005 \quad (11)$$

It seems that the use of (10) in the context of 1D and 2D modelling of wood, is sufficient for fitting the experimental results (Toratti and Svensson 2002; Hanhijärvi and Mackenzie-Helnwein 2003). In the present work, this model is used for describing hygroexpansion in the cross grain direction. For the case of timber beams subjected to bending, however, the following expression for the hygroexpansion strain rate component in longitudinal direction is used:

$$\dot{\epsilon}_L^u = \alpha_L \left(1 - \frac{\beta}{\alpha_L} \epsilon_L^{mech} \right) \dot{u} \quad (12)$$

where ϵ_L^{mech} is the corresponding component of the mechanical strain vector (which describes total strain excluding hygroexpansion). This equation, proposed in Toratti (1992), was found to describe better the response in the grain direction of wood under compressive, tensile and bending loads. Tensile loads resulted in smaller hygroexpansion strains compared to compression loads. This formulation was only tested at low to moderate load levels, as the interest was on working stress levels of structures. A similar approach was used in Hanhijärvi (1995) where the dependence on stress of (12) is also discussed.

2.4 Viscoelastic creep

The additional deformation shown by wood under sustained loads in constant moisture conditions is known as viscoelastic creep. The derivative of the Helmholtz free energy in (2) with respect to the elemental viscoelastic strain provides the thermodynamic stress in the i -th Kelvin element (Hanhijärvi and Mackenzie-Helnwein 2003). Using (5) the following equation holds:

$$\boldsymbol{\sigma}_i^{ve} := - \frac{\partial \Psi}{\partial \boldsymbol{\epsilon}_i^{ve}} = \boldsymbol{\sigma} - \mathbf{C}_i^{ve} : \boldsymbol{\epsilon}_i^{ve} \quad (13)$$

Table 1 Dimensionless compliance parameters used for the viscoelastic model

i	τ_i [h]	J_i^{ve} [–]
1	2.4	0.085
2	24	0.035
3	240	0.07
4	2400	0.2

and the rate equation for a single viscoelastic element can be written as

$$\dot{\boldsymbol{\varepsilon}}_i^{ve} + \frac{1}{\tau_i} \boldsymbol{\varepsilon}_i^{ve} = \frac{1}{\tau_i} \mathbf{C}_i^{ve-1} : \boldsymbol{\sigma} \quad (14)$$

where τ_i represents the retardation (or characteristic) time of the Kelvin element (Malvern 1969; Maugin 1992). In the present work, the elemental compliance viscoelastic tensor is assumed to be proportional to the elastic compliance in the reference conditions:

$$\mathbf{C}_i^{ve-1} = J_i^{ve} \mathbf{C}^{e-1}(u_{ref}) \quad (15)$$

where J_i^{ve} represents a dimensionless viscoelastic compliance ratio compared to the elastic compliance for the i -th element and $u_{ref} = 0.12$. In this paper the dimensionless compliance ratios are calculated on the basis of the viscoelastic model proposed by Toratti (1992) for wood in the grain direction during constant moisture conditions. The use of the same ratios in R and T directions is a simplification of the model. Four Kelvin elements are considered (the values of retardation times and compliances are reported in Table 1). These parameters are used in all moisture conditions for the cases studied in this paper.

As shown in Hanhijärvi and Mackenzie-Helnwein (2003), for stress driven problems the solution of (14) can be written in the following form:

$$\boldsymbol{\varepsilon}_i^{ve} = \boldsymbol{\varepsilon}_{i,n}^{ve} \exp\left(\frac{-\Delta t}{\tau_i}\right) + \int_{t_n}^{t_{n+1}} \frac{\mathbf{C}_i^{ve-1} : \boldsymbol{\sigma}(t)}{\tau_i} \exp\left(-\frac{t_{n+1}-t}{\tau_i}\right) dt \quad (16)$$

where $\Delta t = t_{n+1} - t_n$ is the time increment and $\boldsymbol{\varepsilon}_{i,n}^{ve}$ represents the elemental viscoelastic strain tensor calculated at the time t_n .

2.5 Mechanosorptive creep

The phenomenon of deformation increase in the presence of changing moisture under mechanical loads is known as mechanosorption. For describing recoverable mechanosorption, an equation similar to that of the simple viscoelastic creep is used:

$$\dot{\boldsymbol{\varepsilon}}_j^{ms} = \frac{\mathbf{C}_j^{ms-1} : \boldsymbol{\sigma} - \boldsymbol{\varepsilon}_j^{ms}}{\tau_j} |\dot{u}| \quad (17)$$

where τ_j is a material parameter comparable to a retardation time in viscoelasticity (Hanhijärvi and Mackenzie-Helnwein 2003). This parameter is dimensionless and independent of moisture (see Toratti and Svensson 2002). The solution of (17) is of the same form as (16) but moisture content u is used instead of time:

$$\boldsymbol{\varepsilon}_j^{ms} = \boldsymbol{\varepsilon}_{j,n}^{ms} \exp\left(\frac{-\Delta u^a}{\tau_j}\right) + \int_{u_n}^{u_{n+1}} \frac{\mathbf{C}_j^{ms-1} : \boldsymbol{\sigma}(t)}{\tau_j} \exp\left(-\frac{u_{n+1}-u^a(t)}{\tau_j}\right) dt \quad (18)$$

where $\Delta u = u_{n+1}^a - u_n^a$ with $u^a = |u|$ and $\boldsymbol{\varepsilon}_{j,n}^{ms}$ represents the elemental mechanosorptive strain tensor calculated at the time t_n . The elemental mechanosorptive compliance tensor in matrix form is assumed to be the following:

$$\mathbf{C}_j^{ms-1} = \begin{bmatrix} \frac{m_j^{ms,T} E_T}{E_R} & -m_j^{ms,T} \nu_{TR} & -\frac{m_j^{ms,T} \nu_{LR} E_T}{E_L} & 0 & 0 & 0 \\ -\frac{m_j^{ms,T} \nu_{RT} E_T}{E_R} & m_j^{ms,T} & -\frac{m_j^{ms,T} \nu_{LT} E_T}{E_L} & 0 & 0 & 0 \\ -\frac{m_j^{ms,T} \nu_{RL} E_T}{E_R} & -m_j^{ms,T} \nu_{TL} & m_j^{ms,L} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{m_j^{ms,T} E_T}{G_{RT}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{m_j^{ms,T} E_T}{G_{RL}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{m_j^{ms,T} E_T}{G_{TL}} \end{bmatrix} \quad (19)$$

The coefficients of this matrix are based on: (a) the model for mechanosorption in cross grain section proposed by Toratti and Svensson (2002) and (b) the mechanosorptive model for wood parallel to grain introduced by Toratti (1992). Starting from the mechanosorptive compliance in tangential direction $m_j^{ms,T}$ the other coefficients of $\mathbf{C}_j^{ms-1}$ are obtained by using the ratios between the respective terms of the elastic matrix. However, term $m_j^{ms,L}$ represents a fraction of a reference elastic compliance in longitudinal direction defined as

$$m_0^{ms,L} = \frac{0.7}{E_L(u_{0,ref})} \text{ [1/MPa]} \quad (20)$$

where $u_{0,ref} = 0.20$.

As in the 1D model of Toratti and Svensson (2002), the mechanism for recoverable mechanosorption is composed of three Kelvin elements. Since the full constitutive model presented in the present work is three-dimensional and the influence of the viscoelastic creep is taken into account, the values of the tangential compliance are found to be different from those of the previous 1D model. The same mechanosorptive compliances are used for analyzing all tests described in this work. The values of these compliances and the corresponding retardation times are listed in Table 2.

Following Toratti and Svensson (2002), a further mechanism for describing irrecoverable mechanosorption is added and the expression for the irrecoverable rate is assumed to be

$$\dot{\boldsymbol{\varepsilon}}^{ms,irr} = \mathbf{C}^{ms,irr-1} : \boldsymbol{\sigma} |\dot{U}| \quad (21)$$

Table 2 Tangential and longitudinal compliance parameters for the recoverable mechanosorptive part of the model

i	τ_i [–]	$m_j^{ms,T}$ [1/MPa]	$m_j^{ms,L}$ [1/MPa]
1	0.01	0.0006	$0.25 m_0^{ms,L}$
2	0.1	0.0006	$0.7 m_0^{ms,L}$
3	1.0	0.005	$0.05 m_0^{ms,L}$

where the moisture content rate $\dot{U}$ refers to the moisture levels not reached during the previous strain history. The irrecoverable mechanosorptive matrix is assumed to be the following:

$$\mathbf{C}^{ms,irr-1} = \begin{bmatrix} m_v \frac{E_T}{E_R} & -m_v \frac{E_T}{E_R} \nu_{RT} & 0 & 0 & 0 & 0 \\ -m_v \frac{E_T}{E_R} \nu_{RT} & m_v & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_v \frac{E_T}{G_{RT}} & 0 & 0 \\ 0 & 0 & 0 & 0 & m_v \frac{E_T}{G_{RL}} & 0 \\ 0 & 0 & 0 & 0 & 0 & m_v \frac{E_T}{G_{TL}} \end{bmatrix} \quad (22)$$

where the value of the irrecoverable coefficient m_v in tangential direction, obtained by fitting the experimental results, is $m_v = 0.066 \text{ MPa}^{-1}$. The other coefficients of $\mathbf{C}^{ms,irr-1}$ are calculated from m_v by using the elastic matrix ratios. Equation (21) shows that the deformation due to irrecoverable mechanosorption is considered to be stress rate independent, that is, at a rheological level a mechanosorptive dashpot scheme is used (Hanhijärvi 1995). In some earlier studies mechanosorptive dashpot schemes for describing the full mechanosorptive phenomenon were used (Santaoja et al. 1991; Ormarsson 1999). In the presence of elevated temperatures which may introduce relevant irrecoverable deformations, this approach may not be anymore sufficient and additional Kelvin plastic elements are needed in the rheological scheme (Hanhijärvi and Mackenzie-Helnwein 2003).

3 Coupled moisture-stress analysis by the FEM code Abaqus

3.1 Moisture transfer in wood

In the present work, wood material is assumed to follow Fick's law for moisture transfer and the temperature is considered to be constant. As suggested by Ranta-Maunus (2003), the effect of moisture is considered as a further action to be added to the external mechanical loads. For this reason, a coupled moisture-stress analysis seems to be a suitable computational tool for analyzing the moisture induced stresses in wood. Let Ω be the wood domain, Fick's law is expressed as

$$\left. \frac{\partial u}{\partial t} \right|_{\Omega} = \nabla \cdot (\mathbf{D} \cdot \nabla u) \quad (23)$$

where u is the moisture content of wood and $\mathbf{D}$ the second-order diffusion tensor of moisture transfer. Note that this form of the Fick equation can be used when the density of wood ρ is constant. For varying ρ , the variable $c = \rho u$ instead of u has to be considered as the potential (where c represents the concentration in kg/m^3). In literature, the assumption of isotropic diffusion is usually made and only the diagonal coefficients of the matrix form of tensor $\mathbf{D}$ are considered to be nonzero. As pointed out in Hanhijärvi (1995), the dependence of the diffusion tensor on moisture content makes (23) nonlinear but, since the changes of both u and $D(u)$ are small, $D(u)$ may be considered constant during the single time step and a linear solution of (23) can be accepted.

For a more realistic description of diffusion in 3D wood structures, the values of $D(u)$ in the three material directions should be defined. In Sjödin (2004), the diffusion coefficients in longitudinal direction D_L and in the transversal direction D_T were calculated for Norway

spruce. For the radial direction, $D_R = D_T$ was assumed. In the present work, the following expression of the diffusion coefficient depending on the moisture content is used for wood in radial and tangential direction:

$$D_R(u) = D_T(u) = 8.64 \times 10^{-7} e^{4u} \text{ m}^2\text{h}^{-1} \quad (24)$$

This coefficient, based on the expression provided by Toratti (1992), is obtained by fitting the experimental results presented in Jönsson (2005). In the longitudinal direction, the experimental values of $D_L(u)$ presented by Sjödin (2004) are used. Note that Fick's equation is mathematically analogous to the heat transfer equation:

$$\rho c_T \frac{\partial T}{\partial t} \bigg|_{\Omega} = \nabla \cdot (\boldsymbol{\lambda} \cdot \nabla T) \quad (25)$$

where ρ , c_T and T are the density, the specific heat and the temperature of wood, respectively, while $\boldsymbol{\lambda}$ represents the second-order thermal conductivity tensor. The analogy between (23) and (25) is obtained by posing $c_T = 1$ and $\boldsymbol{\lambda} = \rho \mathbf{D}$. Because of this analogy, the moisture-stress analysis for wood can be performed by using the available temperature-displacement analysis of the Abaqus/Standard code. This code utilizes the UMAT subroutine where the above constitutive model is implemented while the moisture flow equation through the wood surface is implemented into the user subroutine DFLUX. The moisture flow at the surface Γ of the wood body is expressed by using the following equation by Rosen (1978) and Avramidis and Siau (1987):

$$\frac{q_n}{\rho} = S_u(u_{air} - u_{surf}) \quad (26)$$

where q_n represents the moisture flow across the boundary, $S_u = 3.2 \times 10^{-8} e^{4u}$ m/s is the coefficient of surface emission (Hanhijärvi 1995), u_{surf} is the moisture content of the wood surface and u_{air} the equilibrium moisture content of wood corresponding to the air humidity which, according to Avramidis (1989), may be described with the following function:

$$u_{air} = 0.01 \left(\frac{-(T + 273.15) \ln(1 - h)}{0.13(1 - (T + 273.15)/647.1)^{6.46}} \right)^{1/[110(T + 273.15)^{-0.75}]} \quad (27)$$

In (26), u_{surf} is the current solution in terms of moisture content, T represents the temperature in Celsius and $h = 0.01 \cdot \%RH$ where RH is the relative humidity. As pointed out in Hanhijärvi (1995), (27) was obtained in Avramidis (1989) for the use of the sorption isotherm by Zuritz et al. (1979).

In the subroutine DFLUX, both the flow magnitude and the rate of change of flow with respect to the current moisture content are calculated.

3.2 An algorithm for moisture-induced stress update

From a computational point of view, the problem of creep in wood can be solved by using an incremental analysis during time which takes into account the coupling of moisture changes and stresses. The solution in terms of strain, stress and moisture at a given time $t = t_n$ is assumed to be known while the solution at the current time $t = t_{n+1}$ has to be determined. For each deformation mechanism the algorithmic tangent operator, based on the correspondent numerical integration scheme, is built as done in Mackenzie-Helnwein and Hanhijärvi (2003). The single operators are then assembled in a total algorithmic tangent operator.

Let $\Delta \epsilon_{n+1}$ be the total strain increment at time $t = t_{n+1}$ and $\Delta t = t_{n+1} - t_n$ the time increment. Equation (1) can be rewritten in terms of finite increments as

$$\Delta \epsilon_{n+1} = \Delta \epsilon_{n+1}^e + \Delta \epsilon_{n+1}^u + \sum_{i=1}^p \Delta \epsilon_{i,n+1}^{ve} + \sum_{j=1}^q \Delta \epsilon_{j,n+1}^{ms} + \Delta \epsilon_{n+1}^{ms,irr} \quad (28)$$

The total strain increment $\Delta \epsilon_{n+1}$ is composed of a part due to hygroexpansion ($\Delta \epsilon_{n+1}^u$) and of two further parts: one dependent on the stress state at the beginning of the time increment ($\Delta \epsilon_{n+1}^{\sigma_n}$) and the other being a function of the time increment ($\Delta \epsilon_{n+1}^{\Delta t}$). For the model presented in this work, these two parts of the strain increment are:

$$\Delta \epsilon_{n+1}^{\sigma_n} = \Delta \epsilon_{n+1}^e + \epsilon_{n+1}^{ms,irr} \quad (29)$$

$$\Delta \epsilon_{n+1}^{\Delta t} = \Delta \epsilon_{n+1}^e + \sum_{i=1}^p \Delta \epsilon_{i,n+1}^{ve} + \sum_{j=1}^q \Delta \epsilon_{j,n+1}^{ms} \quad (30)$$

The strain increments of (28) as well as the total algorithmic operator of the constitutive model are derived in [Appendix](#) at the end of this paper.

3.3 Implementation into the UMAT subroutine of Abaqus

In this work the algorithm for moisture induced stress update is implemented into the user subroutine UMAT of Abaqus code.

At the beginning of the current time step t_{n+1} , the moisture content u_{n+1} , the moisture increment Δu_{n+1} and the total strain increment $\Delta \epsilon_{n+1}$ have the values calculated by the moisture-stress analysis performed by the program Abaqus/Standard. The value of the strain increment $\Delta \epsilon_{n+1}$ is given by the variable DSTRAN of the UMAT subroutine. The values of total stress, elemental viscoelastic strain and elemental mechanosorptive strain at the beginning of the current step are equal to those provided by the UMAT subroutine at the end of the previous time step t_n . In particular, the value of the total stress is given by the UMAT variable STRESS while the values of creep strain increments are stored into the UMAT array STATEV.

For sake of generality, at the beginning of the current time step, the stress is indicated as $\sigma_{n+1}^{(k=0)}$, the viscoelastic strain $\epsilon_{i,n+1}^{ve(k=0)}$ and the mechanosorptive strain $\epsilon_{j,n+1}^{ms(k=0)}$. Then, the algorithm at the current time step t_{n+1} can be schematically listed in the following way:

1. The elastic compliance matrix $\mathbf{C}^{e-1}$, the irrecoverable mechanosorptive matrix $\mathbf{C}^{ms,irr-1}$ and the strain increments $\Delta \epsilon_{n+1}^{\sigma_n}$ and $\Delta \epsilon_{n+1}^u$ are computed.
2. The elemental algorithmic tangent operators for viscoelastic creep and mechanosorption $\mathbf{C}_{i,n+1}^{ve}$, $\mathbf{C}_{j,n+1}^{ms}$ (based on the elemental matrices $\mathbf{C}_i^{ve-1}$, $\mathbf{C}_j^{ms-1}$) and the tangent operator of the model $\mathbf{C}^T$ are calculated (see [Appendix](#)). This last operator will be used by Abaqus/Standard during the moisture-stress analysis.
3. The new stress increment $\Delta \sigma_{n+1}$ and the creep strain increments $\Delta \epsilon_{n+1}^{ve}$ and $\Delta \epsilon_{n+1}^{ms}$ are calculated (see [Appendix](#)).
4. The updated values of total stress, viscoelastic strain and recoverable mechanosorptive strain are evaluated as

$$\sigma_{n+1}^{(k+1)} = \sigma_{n+1}^{(k)} + \Delta \sigma_{n+1} \quad (31)$$

$$\epsilon_{i,n+1}^{ve(k+1)} = \epsilon_{i,n+1}^{ve(k)} + \Delta \epsilon_{i,n+1}^{ve} \quad (32)$$

$$\boldsymbol{\varepsilon}_{j,n+1}^{ms(k+1)} = \boldsymbol{\varepsilon}_{j,n+1}^{ms(k)} + \Delta \boldsymbol{\varepsilon}_{j,n+1}^{ms} \quad (33)$$

- In the presence of strong nonlinearities and for more complex constitutive models, the scheme can be repeated from step 3 until $\|\mathbf{R}_{n+1}^{(k+1)}\| < TOL$, where $\mathbf{R}_{n+1}^{(k+1)}$ is a vector having as components the residual vectors coming from new forms of (35), (42) and (45) shown in Appendix and TOL is a suitable tolerance. This approach was used in Mackenzie-Helnwein and Hanhijärvi (2003) where a plastic Kelvin element was also introduced for cases of drying under elevated temperatures.
5. The final values of $\boldsymbol{\varepsilon}_{i,n+1}^{ve(k+1)}$ and $\boldsymbol{\varepsilon}_{j,n+1}^{ms(k+1)}$ are stored in the array STATEV of the UMAT subroutine and used for calculating the new quantities of the next time step.

3.3.1 Extension of the method for evaluation of hygrothermal stresses by Abaqus

The algorithm described above can be directly used in the context of a more general analysis for evaluating the stresses induced by both temperature and moisture changes in cases where the influence of temperature is important as, for example, the case of high-temperature drying modelling. This kind of analysis can be performed with Abaqus by using a sequentially coupled method aimed to model both heat and moisture transfer and the induced stresses. The sequential analysis is basically performed in two steps:

1. A transient heat transfer analysis is conducted in order to simulate the temperature field, which is independent of the moisture field;
2. The coupled moisture-stress analysis described in the present work is performed but this time the mechanical quantities are dependent also on the temperature field calculated at step 1.

In the Abaqus code, the results of the first analysis can be easily read by writing suitable instructions into the input data file of the second analysis. This approach will be presented in detail in a further joint paper by the present authors.

4 Validation of the model

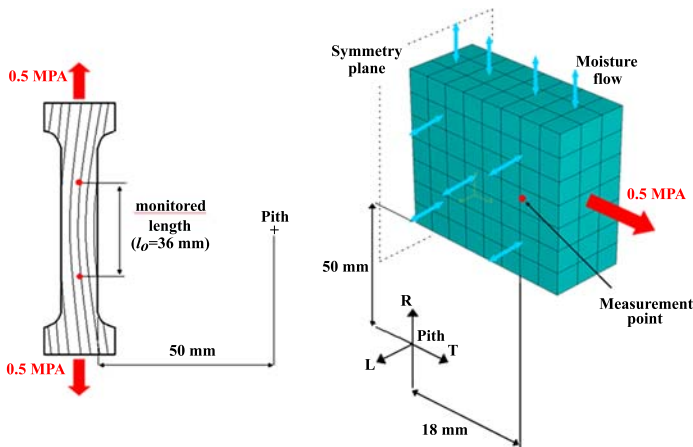
An important part of this present paper is the validation of the presented 3D model for wood by comparisons with experimental results. The experimental data refer to small size timber specimens. The aim of this numerical work is to verify if the model is suitable for simulating both short-term and long-term deformations and stresses in the presence of moisture variations.

The geometrical models are created by means of the CAE module of Abaqus, Version 6.5-3 (Abaqus 2004a). A local cylindrical coordinate system for wood is given and a cylindrical material orientation is assigned (André 2007). The examples studied in the present paper concern straight wood members. Instead, in the case of curved beams, since the piths of the wood parts run along a series of circular arcs, a further Abaqus user subroutine, ORIENT, has to be used in order to define the local material directions (Abaqus 2004a; Ormarsson 1999; Vidal-Sallé and Chassagne 2007).

The simulations performed in this work are carried out by using the Abaqus/Standard program. To mesh the wood parts, 3D coupled elements available for the temperature–displacement type analyses, i.e. the linear C3D8T or the quadratic C3D20T, are used.

Table 3 Material properties for Scots Pine (*Pinus sylvestris*) used in the analysis

E_R (MPa)	E_T (MPa)	E_L (MPa)	ν_{RT}	ν_{RL}	ν_{TL}	G_{RT} (MPa)	G_{RL} (MPa)	G_{TL} (MPa)
900	500	12000	0.558	0.038	0.015	40	700	700


Fig. 3 Test Toratti and Svensson (2000, 2002). *Left*: geometry of the specimen. *Right*: Abaqus model and material coordinate system

4.1 Comparisons with the Toratti and Svensson results (Toratti and Svensson 2000, 2002) in the direction perpendicular to grain

The reference experimental data used in this work were obtained by Toratti and Svensson (2000, 2002) by using clear heartwood samples ($10 \times 20 \text{ mm}^2$ cross-section) of Scots pine (*Pinus sylvestris*). Before testing, the specimens were conditioned at room temperature ($T = 20^\circ\text{C}$) and relative humidity $\text{RH} = 60\%$ for 1 year. Several tests under different humidity cycles were performed. In each test, six of the specimens were used for constant compressive load tests, five for constant tensile load tests and one as a free shrinkage reference. Both the loading and the measuring direction coincide with the tangential direction of the wood for all the studied specimens. The free shrinkage strain was subtracted from both the tensile and the compressive strain results (shrinkage-corrected strain). The values of the material properties used for the analysis are reported in Table 3.

For the numerical simulation, 168 quadratic elements C3D20T are used. In Fig. 3 the scheme of the Toratti and Svensson specimen for the constant tensile load tests and the specimen model analyzed by Abaqus are shown.

4.1.1 Short-term analysis

These tests were carried out at different climate cycles (RH versus time) in both high humidity ($\text{RH} = 60\text{--}90\%$) and low humidity ($\text{RH} = 60\text{--}45\%$) conditions. Fast humidity cycle (step-cycle) and slow humidity cycles (triangular cycle) were tested. The specimens were loaded by a moderate tensile or compressive stress of 0.5 MPa and unloaded after 168 hours.

Single humidity cycles were applied. During the loading time, a first humidity cycle was imposed for checking its effect on the strain distribution. After unloading, the second humidity cycle was imposed for an other week in order to study the induced recovery. The experimental results for the tension case showed that the deformations during wetting are mostly influenced by the mechanosorptive effect with respect to those during drying, where this effect is not evident. On the contrary, in the compression case, the influence of mechanosorption can be noticed also during drying. Furthermore, the induced deformations were found to be only partially recovered after unloading. Figure 4 shows the comparisons between computational and experimental shrinkage-corrected strains and relative creep curves in case of high humidity cycles for short-term analysis and tensile load test with step-cycle humidity. The reported experimental data is the mean value of six tensile test results. The results show that the computational simulation is able to fit well these experimental data considering that the scatter of the original test results was high. Also the comparison with the experimental compressive deformations (mean value of five compressive test results) shows the good performance of the computational model. An exception is however the recovery period after a humidity cycle, in fact the model is not able to fully show the recovery noticed from the experiments.

4.1.2 Long-term analysis

In this case, the experiments were carried out for the same specimens described above in order to study the effect of multiple cycles. The load was applied at the beginning of the test and removed after 1400 hours. The experimental results showed that the measured mechanosorptive strain was about 10 times the elastic strain while for the single humidity cycle this was about five times the elastic strain.

The comparisons between computational and experimental results for the tensile and the compressive case are presented in Fig. 5. Also for this multi-cycle case, shrinkage-corrected strains and relative creep curves are plotted. The results show good performance of the computational method to simulate moisture induced deformations in timber cross sections. The oscillation of the test curves, which is not as clearly shown in the model, is probably due to differences of shrinkage strains, which is highly variable in wood by nature. Most probably not all shrinkage strains were subtracted from the test results curves.

4.2 Comparisons with Leivo's test results (Leivo 1991) for bending wood beams

Leivo (1991) carried out experiments in bending of beams in spruce (*Picea abies*) loaded to a stress of 10 MPa. The member cross section was $45 \times 95 \text{ mm}^2$. Initially the specimens were conditioned at $\text{RH} = 90\%$ and the relative humidity was cycled between $\text{RH} = 35\%$ and $\text{RH} = 90\%$ at a cycle length of 70 days. The starting relative humidity was $\text{RH} = 90\%$. The experiments were carried out in a constant temperature of $T = 21^\circ\text{C}$. The relative creep for eight specimens was measured. The values of material stiffness parameters are reported in Table 4.

The geometry of the beam and the Abaqus model used in the present work are shown in Fig. 6. To model the wood parts, 7207 elements C3D8T are used. Because of the presence of a steel plate attached to the wood beam, 2D shell elements S3 and a rigid body option are used to mesh the plate. The contacts are modeled by a hard contact pair with a penalty method in the tangential direction using a 0.4 penalty factor.

The results presented in Fig. 7 show that the simulated relative creep is within the scatter of the experimental relative creep curves. The computational model seems to perform well also in the grain direction.

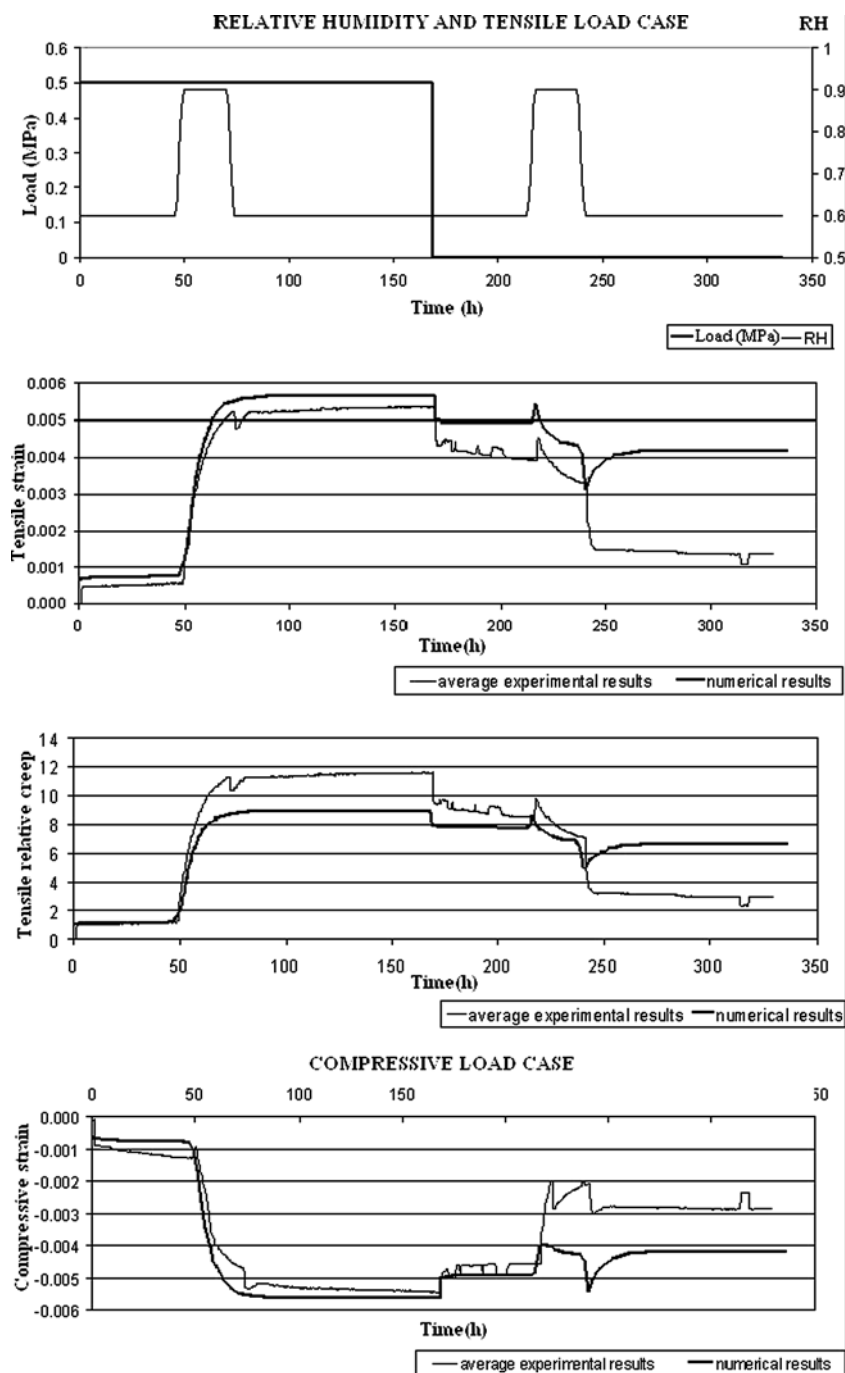


Fig. 4 Test Toratti and Svensson (2000). Short-term analysis. Tension case: mechanical loading, relative humidity, shrinkage-free strains and relative creep. Compression case: strains versus time

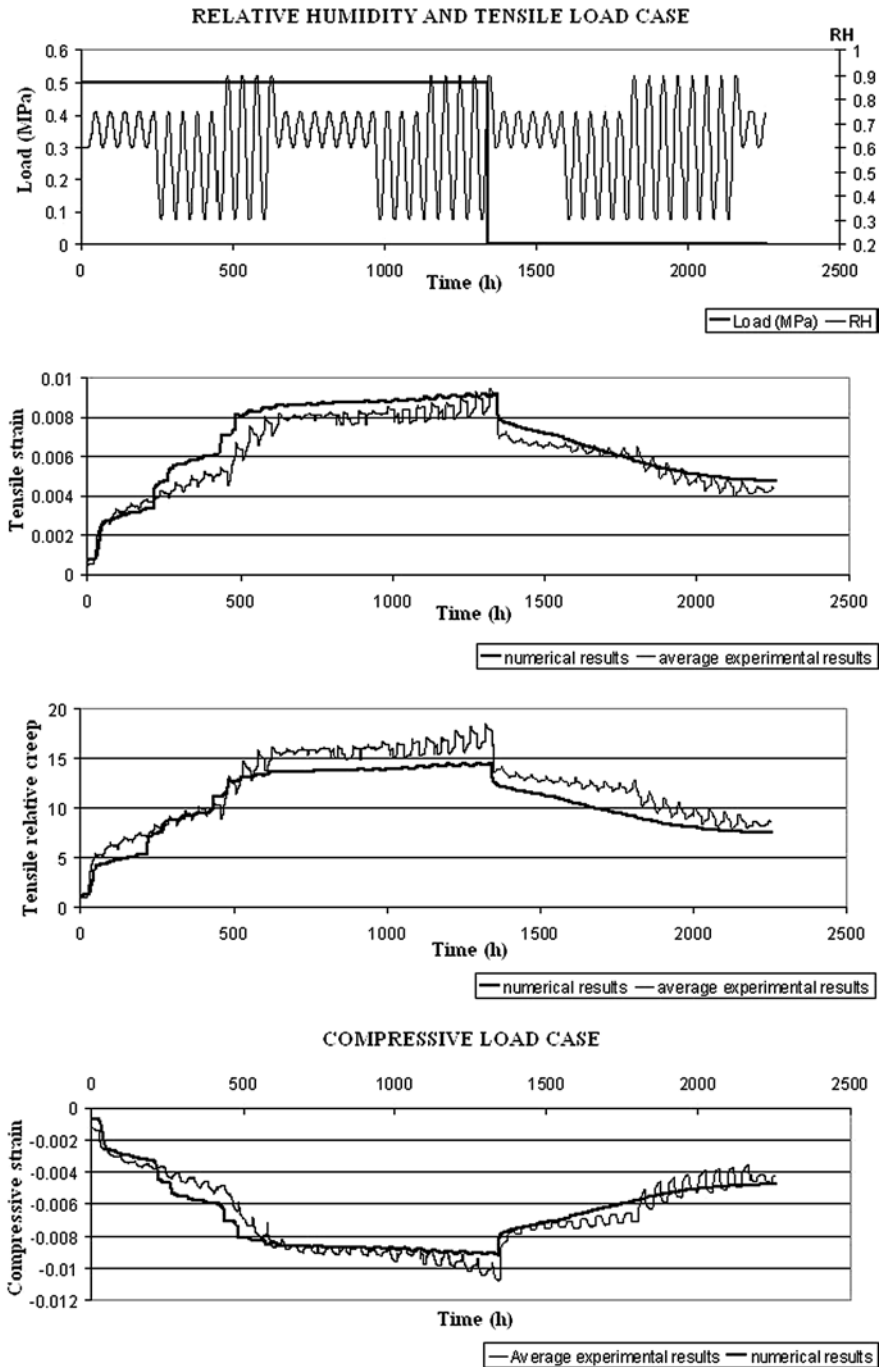
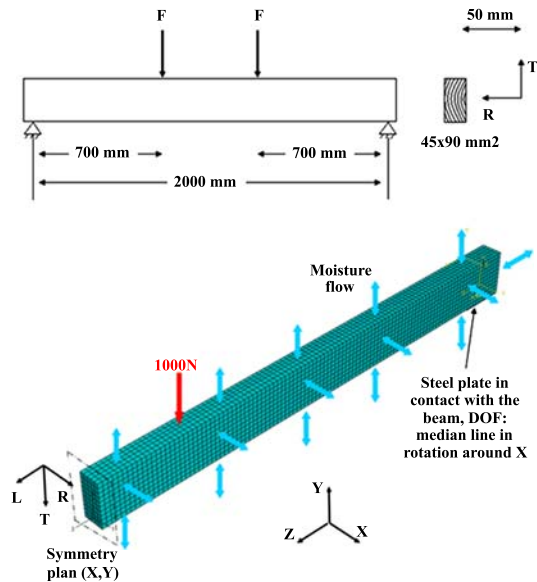


Fig. 5 Test Toratti and Svensson (2002). Long-term analysis. Tension case: mechanical loading, relative humidity, shrinkage-free strains and relative creep. Compression case: strains versus time

Table 4 Mechanical properties for Norway spruce (*Picea abies*) used in the analysis

E_R (MPa)	E_T (MPa)	E_L (MPa)	ν_{RT}	ν_{RL}	ν_{TL}	G_{RT} (MPa)	G_{RL} (MPa)	G_{TL} (MPa)
600	600	12000	0.558	0.038	0.015	40	700	700

Fig. 6 Bending test Leivo (1991). *Top*: geometry of the wood bending beam. *Bottom*: Abaqus model


4.3 Comparisons with Jönsson's test results (Jönsson 2005) for glulam beams

This comparison case is used for studying the internal stresses perpendicular to grain in glued laminated (glulam) timber beams. The described beam section was experimentally analyzed by Jönsson (2005). The tested glulam beams (90 × 270, grade L40) contained six lamellae of Norway spruce (see properties in Table 4). The glulam plate specimen were subjected to an internal moisture gradient in the width direction by altering the ambient air relative humidity from the equilibrium condition of $RH = 40\%$ to $RH = 80\%$. Only the two opposite sides of the specimen were open to air. At different stages, after 3 days and after 38 days of moistening, the specimens were sliced in order to measure the moisture and stress gradients in the piece. The specimen was sawn into 16 mm thick plates perpendicular to the beam axis, so that the knot free specimens had dimensions 16 × 90 × 270 mm. From a single specimen, 11 slices in the width direction were sawn. The length between markers in a slice was measured before and after sawing by an optical method. The mean stress was determined from the mean released strain (ε_{mean}) for each slice:

$$\sigma_{mean} = E(u)\varepsilon_{mean} \quad (34)$$

where the elastic modulus $E(u)$, depending on the moisture content, was measured with a dynamic method.

For the numerical simulation by Abaqus, 396 quadratic elements C3D20T are used. Figure 8 shows the geometry of the Jönsson glulam beam section and the model analyzed by

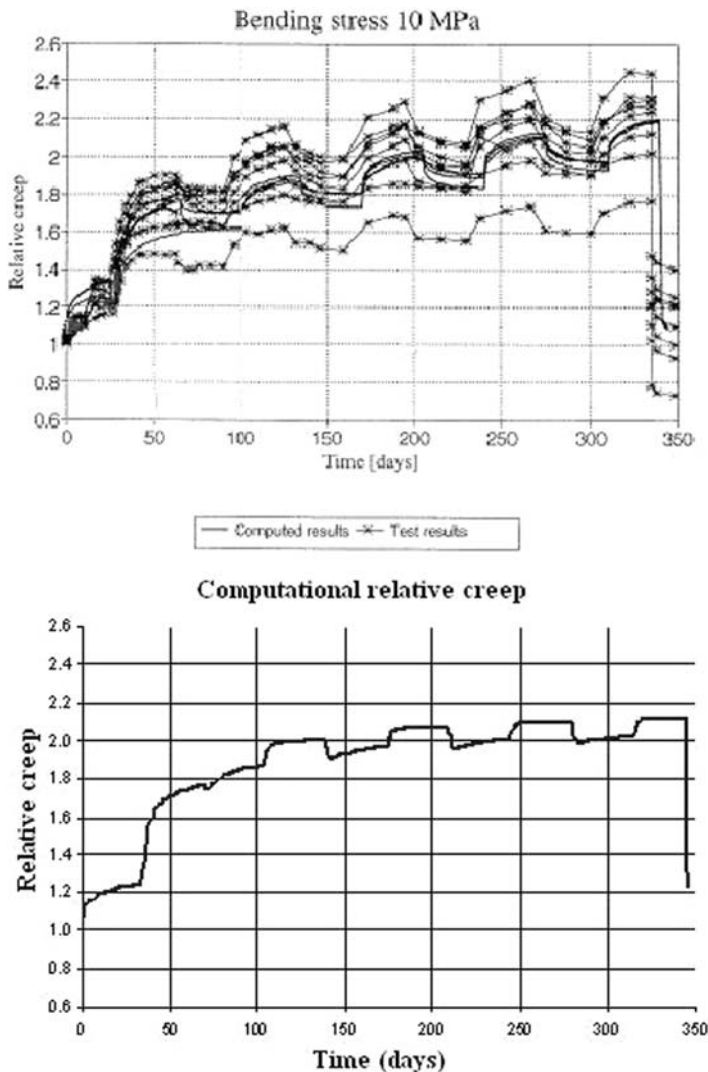


Fig. 7 Bending test Leivo (1991). Wood bending beam. *Top*: experimental and computational relative creep curves from Toratti (1992). *Bottom*: computational results by the computational model presented in this work

Abaqus/Standard. The computational results in terms of moisture content (Fig. 9) and tangential stress (see Fig. 10), show a relatively good agreement with the experimental data. The distribution of stress and moisture are shown in Fig. 11.

5 Conclusions and future work

In this paper, a 3D orthotropic viscoelastic–mechanosorptive model for wood under variable conditions is presented. The constitutive model is based on the 1D model for wood in longitudinal direction introduced by Toratti (1992) and on the 1D model for wood in

Fig. 8 Test Jönsson (2005). *Left*: geometry of the glulam beam. *Right*: Abaqus model (X – Y – Z : global coordinate system)

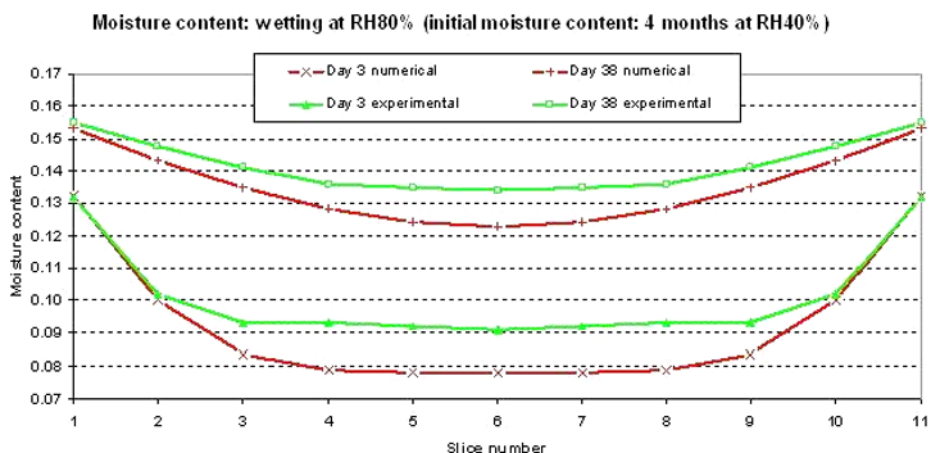
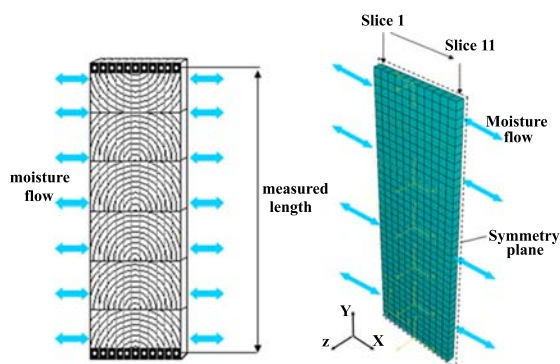


Fig. 9 Test Jönsson (2005). Average of the humidity for each slice. Comparisons between computational and experimental results

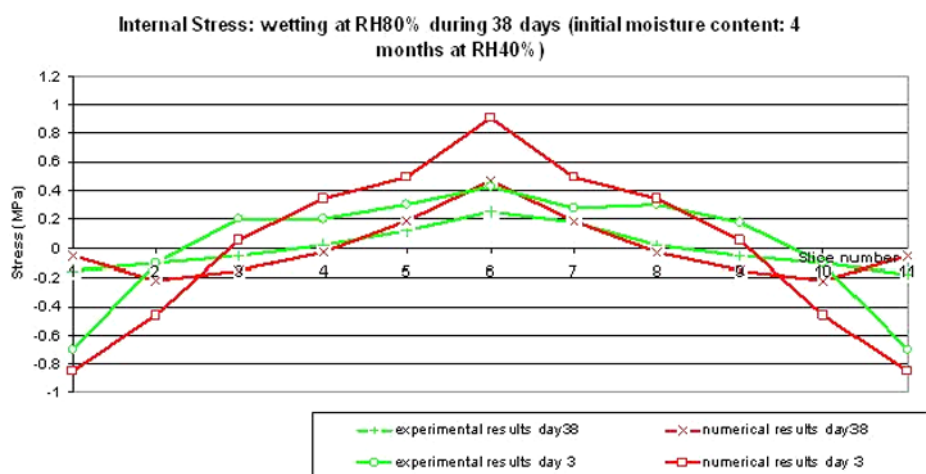
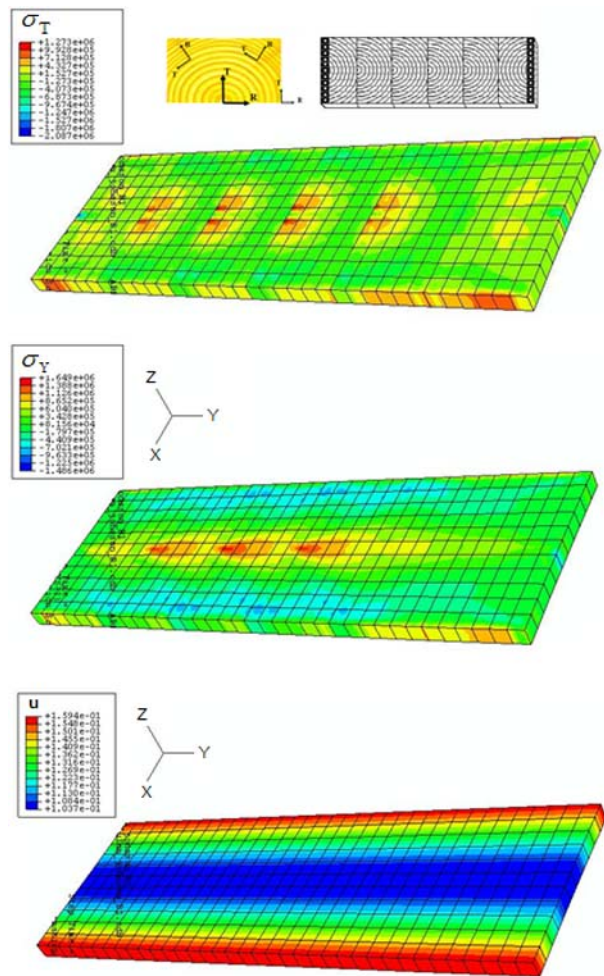


Fig. 10 Test Jönsson (2005). Average of the tangential stress for each slice. Comparisons between computational stresses and experimental mean stresses

Fig. 11 Test Jönsson (2005). Analysis after 4 months and 38 days. *Top*: tangential stresses in material coordinates. *Centre*: tangential stresses in global coordinates. *Bottom*: Moisture content distribution



the perpendicular to grain direction proposed by Toratti and Svensson (2002). For both the viscoelastic and the recoverable mechanosorptive mechanisms, Kelvin elements are used and a mechanosorptive dashpot scheme for irrecoverable mechanosorption is also taken into account. The above creep mechanisms are added to the elastic strain and to the hygroexpansion strain. The whole model is characterized by the fact that the mechanical quantities depend on the moisture content. The temperature is considered to be constant during the studied cases. The extension of the previous 1D constitutive creep models to 3D is done by writing the problem in a thermodynamic form, as proposed by Hanhijärvi and Mackenzie-Helnwein (2003) for a different model used for drying and high temperatures. 3D elemental compliance matrices are defined for each element of the different creep mechanisms and an algorithm for updating the total stress is used which is a variant of the one introduced in Mackenzie-Helnwein and Hanhijärvi (2003). This computational approach is very general and can also be used for other types of materials where the creep mechanism is described by Kelvin elements.

The constitutive model and the algorithm for stress update are implemented into the UMAT subroutine of the FEM code Abaqus. The equation describing the flow across the

surface is implemented into the user subroutine DFLUX. A coupled moisture-stress analysis is performed by using the analogy with the available temperature-displacement analysis of Abaqus.

The validation of the model by comparison with existing experimental data is an important part of the present paper. The computational results fit relatively well the experimental curves for both grain direction and cross grain section. However, more experimental data are needed in order to define viscoelastic and mechanosorptive compliances fully independent of the particular studied case. Modelling of strain recovery needs also further consideration. Besides small solid timber sections, the method appears also suitable for the analysis of glued laminated timber structures. The use of the presented constitutive model for the analysis of larger size timber structures, as for example connections, requires some further experimental work and the comparison of the obtained experimental data with the computational results. This research topic is under development.

Both the material model and the computational method can be directly used for analyzing problems in which the effect of temperature is significant. In this case a thermal analysis and the described coupled moisture-stress analysis can be solved sequentially by using the computational tools of Abaqus. This approach will be presented by the authors in a further work.

The constitutive model and the algorithm for stress update based on elemental algorithmic operators, could be suitable for the calculation of a generalized J integral (Dubois and Petit 2005) in problems of wood fracture mechanics characterized by creep-crack growth in variable climate conditions. This research topic is under study as well.

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Appendix: Computation of the strain increments and of the tangent operator

Elastic strain increment

The elastic strain increment is composed of two parts coming from the finite increment form of the elastic rate equation (9):

$$\Delta \boldsymbol{\varepsilon}_{n+1}^e = \Delta \boldsymbol{\varepsilon}_{n+1}^{e, \sigma_n} + \Delta \boldsymbol{\varepsilon}_{n+1}^{e, \Delta t} \quad (35)$$

with

$$\Delta \boldsymbol{\varepsilon}_{n+1}^{e, \sigma_n} = \dot{\mathbf{C}}^{e-1} : \boldsymbol{\sigma}_n \Delta t \quad (36)$$

$$\Delta \boldsymbol{\varepsilon}_{n+1}^{e, \Delta t} = \mathbf{C}^{e-1} : \dot{\boldsymbol{\sigma}}_{n+1} \Delta t \quad (37)$$

In the used model, the moisture content rate is assumed to have a constant value during each time increment Δt . By using relationships (8) for the elastic moduli in the case of constant density and temperature, the following equation holds (see also Santaoja et al. 1991):

$$\Delta \boldsymbol{\varepsilon}_{n+1}^{e, \sigma_n} = - \frac{c_1 \Delta u_{n+1}}{1 + c_1(u_{n+1} - u_{ref})} \mathbf{C}^{e-1} : \boldsymbol{\sigma}_n \quad (38)$$

where the moisture content increment is defined as $\Delta u_{n+1} = u_{n+1} - u_n$.

A linear relationship is assumed during each time increment so that the increment of the elastic strain depending on Δt at the current time step is

$$\Delta \boldsymbol{\varepsilon}_{n+1}^{e, \Delta t} = \mathbf{C}^{e-1} : \Delta \boldsymbol{\sigma}_{n+1} \quad (39)$$

where $\Delta \boldsymbol{\sigma}_{n+1}$ has to be calculated by the algorithm of moisture induced stress update.

Hygroexpansion strain increment

Starting from (10), the increment of hygroexpansion strain is directly dependent on the moisture increment:

$$\Delta \boldsymbol{\varepsilon}_{n+1}^u = \alpha \Delta u_{n+1} \quad (40)$$

When rate equation (12) is used for describing hygroexpansion in the longitudinal direction, the corresponding increment of strain is evaluated as

$$\Delta \varepsilon_{L, n+1}^u = \alpha_L \left(1 - \frac{\beta}{\alpha_L} \varepsilon_{L, n}^{mech} \right) \Delta u_{n+1} \quad (41)$$

Recoverable creep strain increments

By integrating the equation of viscoelastic creep during the time for the i -th Kelvin element, the following increment of elemental viscoelastic strain is obtained:

$$\Delta \boldsymbol{\varepsilon}_{i, n+1}^{ve} = \mathbf{C}_{i, n+1}^{ve-1} : \Delta \boldsymbol{\sigma}_{n+1} - \mathbf{R}_{i, n}^{ve}(\boldsymbol{\varepsilon}_{i, n}^{ve}, \boldsymbol{\sigma}_n) \quad (42)$$

where $\mathbf{C}_{i, n+1}^{ve}$ is the algorithmic viscoelastic tangent operator:

$$\mathbf{C}_{i, n+1}^{ve} = \frac{\mathbf{C}_i^{ve}}{\mathbb{T}_{n+1}(\xi_i)} \quad (43)$$

with $\xi_i = \frac{\Delta t}{\tau_i}$, $\mathbb{T}_{n+1}(\xi_i) = 1 - \frac{1}{\xi_i} [1 - \exp(-\xi_i)]$ and $\mathbf{R}_{i, n}^{ve}$ is a function of the strain $\boldsymbol{\varepsilon}_{i, n}^{ve}$ and stress $\boldsymbol{\sigma}_n$ at the previous time step (see Mackenzie-Helnwein and Hanhijärvi 2003 for more detail on this computational approach):

$$\mathbf{R}_{i, n}^{ve} = [\boldsymbol{\varepsilon}_{i, n}^{ve} - \mathbf{C}_i^{ve-1} : \boldsymbol{\sigma}_n] [1 - \exp(-\xi_i)] \quad (44)$$

Since the equation for mechanosorption is analogous to that of viscoelastic creep, a similar finite increment for the j -th Kelvin element is obtained:

$$\Delta \boldsymbol{\varepsilon}_{j, n+1}^{ms} = \mathbf{C}_{j, n+1}^{ms-1} : \Delta \boldsymbol{\sigma}_{n+1} - \mathbf{R}_{j, n}^{ms}(\boldsymbol{\varepsilon}_{j, n}^{ms}, \boldsymbol{\sigma}_n) \quad (45)$$

where $\mathbf{C}_{j, n+1}^{ms}$ is the algorithmic mechanosorptive tangent operator:

$$\mathbf{C}_{j, n+1}^{ms} = \frac{\mathbf{C}_j^{ms}}{\mathbb{T}_{n+1}(\xi_j)} \quad (46)$$

being $\xi_j = \frac{\Delta t}{\tau_j}$, $\mathbb{T}_{n+1}(\xi_j) = 1 - \frac{1}{\xi_j} [1 - \exp(-\xi_j)]$ and $\mathbf{R}_{j, n}^{ms}$ a function of $\boldsymbol{\varepsilon}_{j, n}^{ms}$ and $\boldsymbol{\sigma}_n$ at the previous time step:

$$\mathbf{R}_{j, n}^{ms} = [\boldsymbol{\varepsilon}_{j, n}^{ms} - \mathbf{C}_j^{ms-1} : \boldsymbol{\sigma}_n] [1 - \exp(-\xi_j)] \quad (47)$$

Irrecoverable mechanosorptive creep strain increment

The increment due to irrecoverable mechanosorption depends on both the moisture increment and the stress at the beginning of the time step:

$$\Delta \epsilon_{n+1}^{ms,irr} = \mathbf{C}^{ms,irr^{-1}} : \sigma_n |\Delta U| \quad (48)$$

In the above equation, $\Delta U = U_{n+1} - U_n$ with $U_{n+1} = u_{n+1}$ and $U_n = u_{1,n}$ where $u_{1,n}$ represents the value of moisture content u_n not yet reached during the previous strain history. Finally, $\Delta U = 0$ for all U_n already reached.

Tangent operator of the model

Starting from the incremental form of the total strain (28) and taking into account (38), (39), (42) and (45), the following equation holds:

$$\mathbf{C}_T^{-1} : \Delta \sigma_{n+1} = \Delta \epsilon_{n+1} - \Delta \epsilon_{n+1}^{e,\sigma_n} - \Delta \epsilon_{n+1}^u - \Delta \epsilon_{n+1}^{ms,irr} + \sum_{i=1}^p \mathbf{R}_{i,n}^{ve} + \sum_{j=1}^q \mathbf{R}_{j,n}^{ms} \quad (49)$$

where $\mathbf{C}_T$ represents the algorithmic tangent operator of the whole model:

$$\mathbf{C}_T = \left(\mathbf{C}^{e^{-1}} + \sum_{i=1}^p \mathbf{C}_{i,n+1}^{ve^{-1}} + \sum_{j=1}^q \mathbf{C}_{j,n+1}^{ms^{-1}} \right)^{-1} \quad (50)$$

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